Figures

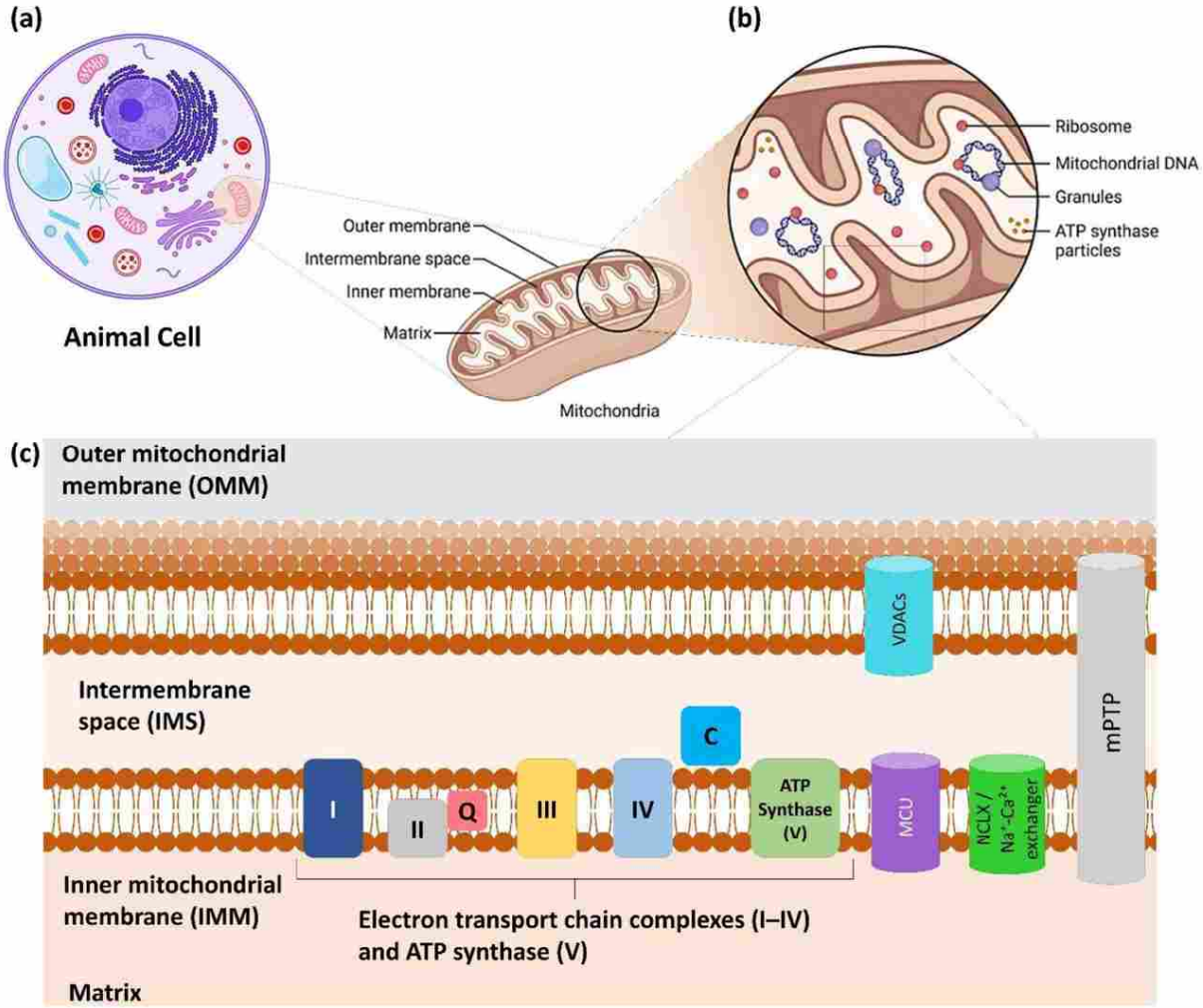

Figure 1. Mitochondrial structure and key functional components. *Panel* ***(a)*** *shows a schematic of an animal cell with mitochondria highlighted among other organelles. Panel* ***(b)*** *provides an enlarged view of mitochondrial ultrastructure, revealing the outer membrane, intermembrane space, inner membrane with cristae folds, and the matrix. Ribosomes, mitochondrial DNA (mtDNA), granules, and ATP synthase particles are also indicated within the organelle. Panel* ***(c)*** *illustrates the detailed molecular organization of the mitochondrial membranes. The outer mitochondrial membrane (OMM) contains voltage dependent anion channels (VDACs), while the inner mitochondrial membrane (IMM) houses the electron transport chain complexes (I to IV), along with the mobile electron carrier coenzyme Q and cytochrome c, and ATP synthase (Complex V). Calcium transport is facilitated by the mitochondrial calcium uniporter (MCU) and the sodium calcium exchanger (NCLX). The mitochondrial permeability transition pore (mPTP) is shown spanning both membranes. Collectively, these components are involved in ATP production, calcium homeostasis, and regulation of apoptosis. Adapted from Frey and Mannella (2000) and Kühlbrandt (2015).*

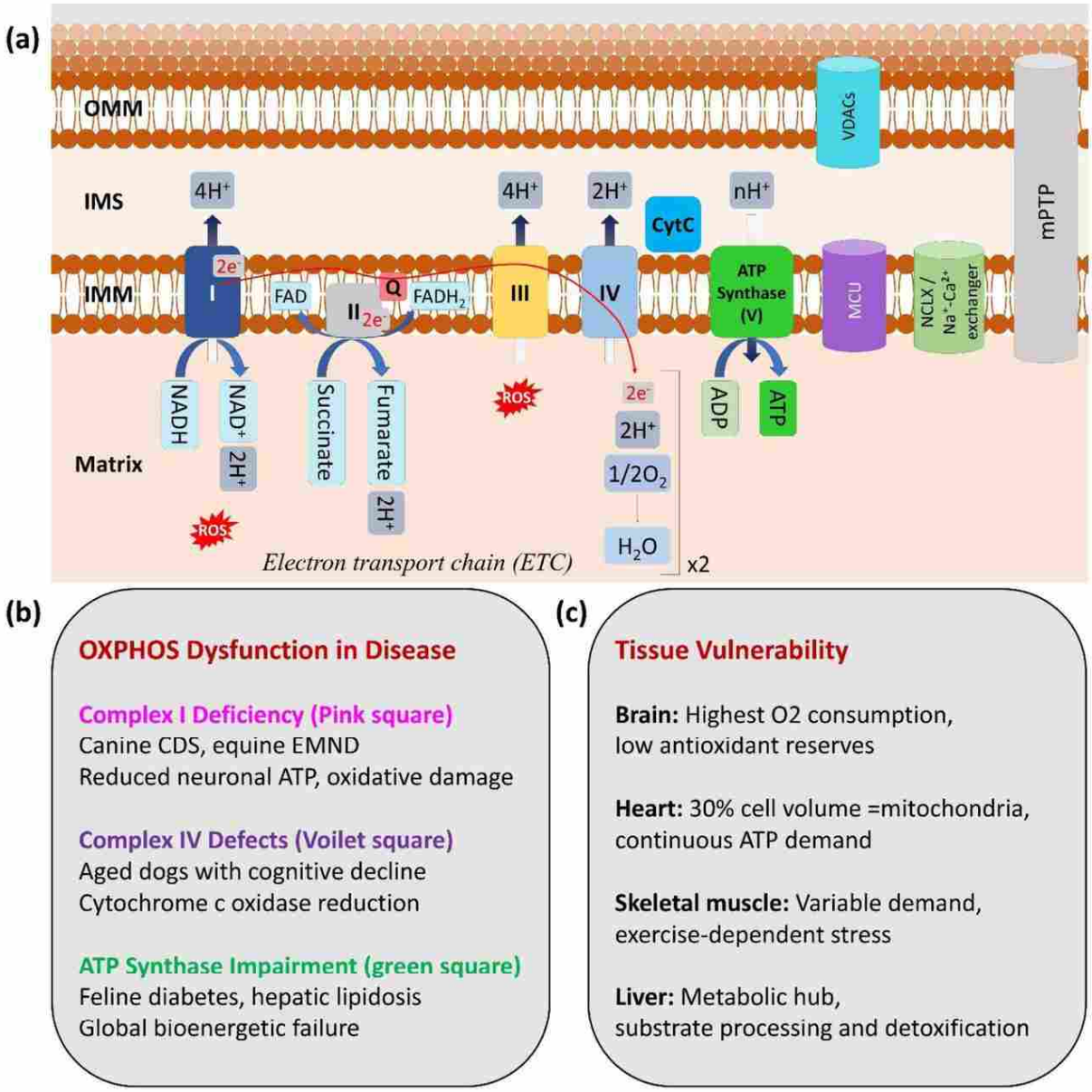

Figure 2. Oxidative phosphorylation and energy failure in animal diseases. *Panel* ***(a)*** *shows the electron transport chain (ETC) within the inner mitochondrial membrane. Complex I accept electrons from NADH and pump four protons into the intermembrane space, while Complex II oxidizes succinate to fumarate. Coenzyme Q and cytochrome c shuttle electrons between the complexes. Complex III pumps four protons and Complex IV transfers electrons to oxygen to form water, pumping two additional protons. This proton gradient drives ATP synthase (Complex V) to generate ATP from ADP. Red burst symbols mark sites of reactive oxygen species (ROS) production near Complexes I and III. Voltage dependent anion channels (VDACs), the mitochondrial calcium uniporter (MCU), sodium calcium exchanger (NCLX), and mitochondrial permeability transition*

601 *pore (mPTP) are also shown. Panel **(b)** outlines OXPHOS dysfunction in veterinary diseases,*
602 *including Complex I deficiency in canine cognitive dysfunction syndrome and equine motor neuron*
603 *disease, Complex IV defects in aged dogs with cognitive decline, and ATP synthase impairment in*
604 *feline diabetes and hepatic lipidosis. Panel **(c)** summarizes tissue vulnerability by metabolic*
605 *demand, covering the brain, heart, skeletal muscle, and liver. Based on Schon et al. (2012) and*
606 *Valberg et al. (2016).*

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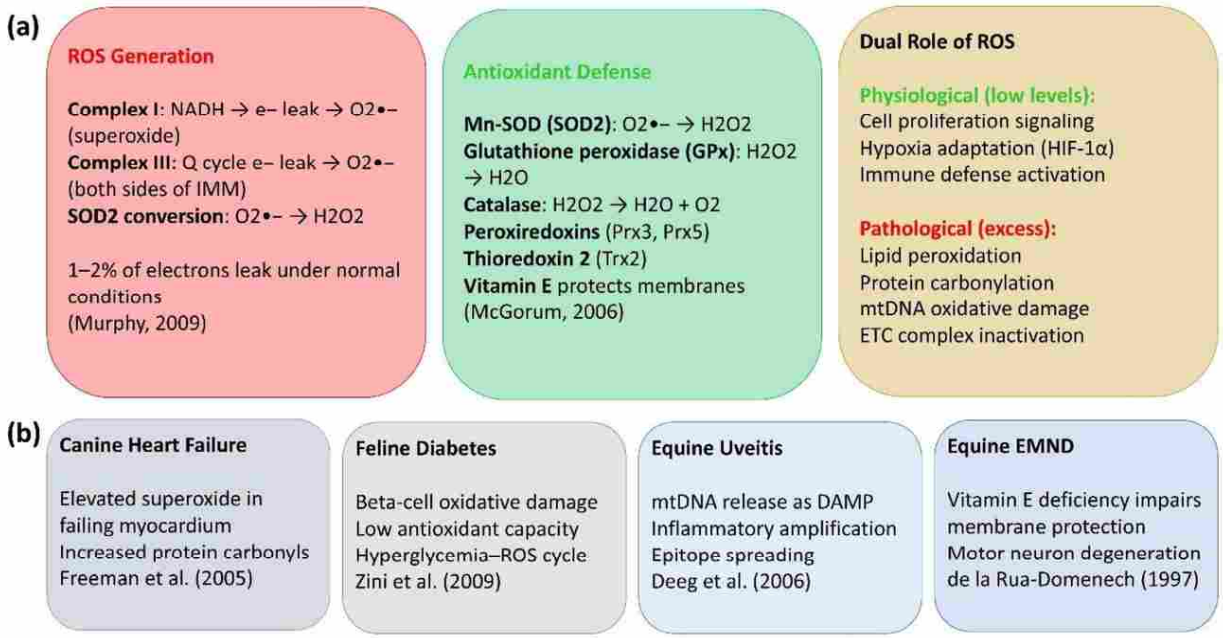

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609 ***Figure 3. Mitochondrial ROS production, defense, and pathological consequences.*** *Panel **(a)***
610 *presents three aspects of mitochondrial reactive oxygen species (ROS) biology. The left box*
611 *describes ROS generation through electron leakage at Complexes I and III, where approximately*
612 *1 to 2 percent of electrons escape under normal conditions and superoxide is converted to*
613 *hydrogen peroxide by SOD2 (Murphy, 2009). The center box outlines antioxidant defense systems*
614 *including manganese superoxide dismutase (Mn SOD), glutathione peroxidase (GPx), catalase,*
615 *peroxiredoxins (Prx3 and Prx5), thioredoxin 2 (Trx2), and vitamin E for membrane protection*
616 *(McGorum, 2006). The right box illustrates the dual role of ROS, with low levels supporting cell*
617 *proliferation, hypoxia adaptation through HIF 1α, and immune defense, while excess ROS causes*
618 *lipid peroxidation, protein carbonylation, mtDNA damage, and ETC complex inactivation. Panel*
619 ***(b)** highlights species specific oxidative stress manifestations in canine heart failure (Freeman et*
620 *al., 2005), feline diabetes (Zini et al., 2009), equine recurrent uveitis (Deeg et al., 2006), and*
621 *equine motor neuron disease (de la Rua Domenech et al., 1997). Based on Murphy (2009) and*
622 *Sena and Chandel (2012).*

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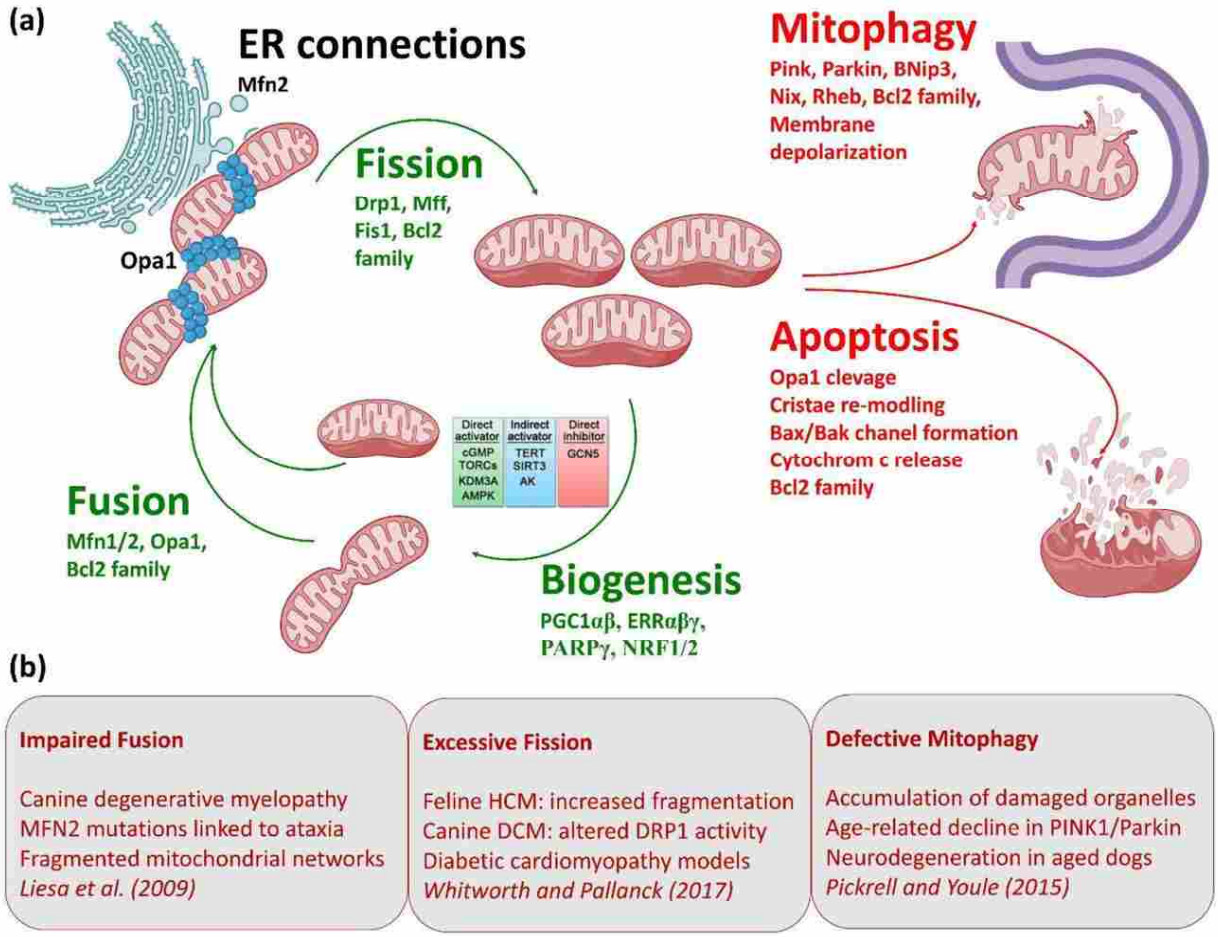

Figure 4. Mitochondrial quality control, dynamics, and mitophagy. *Panel* ***(a)*** *illustrates the major pathways that maintain mitochondrial health. Fusion, carried out by mitofusin 1 and 2 (Mfn1/2), Opa1, and the Bcl2 family, allows partially damaged organelles to complement each other. The endoplasmic reticulum (ER) maintains contact with mitochondria through Mfn2. Fission involves Drp1 along with its receptors Mff and Fis1, enabling the cell to segregate and isolate damaged components. When mitochondria become depolarized, mitophagy is initiated through PINK1 accumulation and Parkin recruitment, with additional roles for BNip3, Nix, Rheb, and the Bcl2 family. The figure also depicts apoptosis, which involves Opa1 cleavage, cristae remodeling, formation of Bax and Bak channels, cytochrome c release, and regulation by the Bcl2 family. A central table outlines the regulation of mitochondrial biogenesis through PGC1αβ, ERRαβγ, PARPγ, and NRF1/2, along with their direct activators (cGMP, TORCδ, KDM3A, AMPK), indirect activators (TERT, SIRT3, AK), and a direct inhibitor (GCN5). Panel* ***(b)*** *connects these processes to diseases, showing impaired fusion in canine degenerative myelopathy due to MFN2 mutations (Liesa et al., 2009), excessive fission in feline hypertrophic cardiomyopathy and canine dilated cardiomyopathy with altered DRP1 activity (Whitworth and Pallanck, 2017), and defective mitophagy with declining PINK1 and Parkin function contributing to neurodegeneration in aged dogs (Pickrell and Youle, 2015). Based on Chan (2012) and Pickrell and Youle (2015).*

642

(a) Comparative Mitochondrial Pathology in Neurodegenerative Diseases

Canine CDS
(cf. Human Alzheimer's)

Pathology:
Amyloid-beta deposition
Reduced Complex IV activity
mtDNA oxidative damage
Altered mitochondrial morphology

Evidence strength:
Strong animal model
Aβ pathology well-characterized

Treatment response:
Antioxidant + enrichment diet
improved cognition in aged dogs

Parkinsonism Models
(Toxin-induced in animals)

Pathology:
Complex I inhibition (MPTP)
Dopaminergic neuron loss
Increased mitochondrial ROS
Impaired mitophagy

Evidence strength:
Toxin models well-validated
Spontaneous PD rare in animals

Controversy:
MPTP models may not fully
replicate human PD pathology

Equine MND
(cf. Human ALS)

Pathology:
Reduced cytochrome c oxidase
Motor neuron degeneration
Vitamin E deficiency
Oxidative membrane damage

Evidence strength:
Nutritional link established;
mechanism not fully elucidated

Unique feature:
Preventable with vitamin E
supplementation (early stages)

(b) Mitochondrial Dysfunction in Cardiovascular Diseases of Companion Animals

Canine DCM

Affected breeds:
Doberman, Boxer, Great Dane

Mitochondrial defects:
Reduced respiratory chain activity
Altered membrane potential
Elevated oxidative stress
~30% cell volume = mitochondria
(note: heart)

Therapeutic targets:
CoQ10 supplementation
L-carnitine support
Metabolic modulators

Feline HCM

Primary cause:
MYBPC3 mutation (sarcomeric)

Secondary mito dysfunction:
Increased energy demand
Calcium overload in matrix
Excessive mitochondrial fission
Oxidative ETC damage

Vicious cycle:
Sarcomeric defect
→ energy demand → mito stress
→ Ca2+ mishandling
→ worsened hypertrophy

Ischemia–Reperfusion

Ischemic phase:
OXPHOS failure,
ATP depletion
Calcium accumulation in matrix
Metabolic intermediate buildup

Reperfusion phase:
Massive ROS burst from ETC
mPTP opening → swelling
Cytochrome c release
Cardiomyocyte death

Protection strategies:
Cyclosporin A (mPTP inhibitor)
MitoQ (targeted antioxidant)
Ischemic preconditioning

643

644 ***Figure 5. (a) Comparative mitochondrial pathology in neurodegenerative diseases across***
645 ***species.*** *Left: canine cognitive dysfunction syndrome (CDS) features amyloid-beta deposition,*
646 *reduced Complex IV activity, mtDNA oxidative damage, and altered morphology, closely*
647 *paralleling human Alzheimer disease; antioxidant-enriched diets improved cognition in aged dogs*
648 *(Milgram et al., 2005). Center: toxin-induced parkinsonism models (MPTP, rotenone) in dogs and*
649 *primates show Complex I inhibition and dopaminergic neuron loss, though spontaneous PD is rare*
650 *in domestic animals; a noted controversy is that toxin models may not replicate the progressive*
651 *nature of idiopathic PD. Right: equine motor neuron disease (EMND) shows reduced cytochrome*
652 *c oxidase activity and oxidative membrane damage linked to vitamin E deficiency, with the unique*
653 *feature of being preventable through supplementation. Evidence strength bars indicate relative*
654 *robustness of the supporting data for each condition.* ***(b) Mitochondrial dysfunction in***
655 ***cardiovascular diseases of companion animals.*** *Left: canine dib) Mitochondrialthy (DCM) in*
656 *Doberman Pinschers, Boxers, and Great Danes, showing reduced respiratory chain activity,*

altered membrane potential, and oxidative stress in cardiomyocytes where mitochondria occupy approximately 30% of cell volume; CoQ10 and L-carnitine supplementation used therapeutically. Center: feline hypertrophic cardiomyopathy (HCM) driven by MYBPC3 sarcomeric mutations, with secondary mitochondrial dysfunction creating a vicious cycle of energy demand, calcium overload, and worsened hypertrophy. Right: ischemia-reperfusion injury showing sequential ischemic (ATP depletion, calcium accumulation) and reperfusion (ROS burst, mPTP opening) phases, with protective strategies including cyclosporin A, MitoQ, and ischemic preconditioning. Based on Brown et al. (2017) and Maron and Fox (2015).

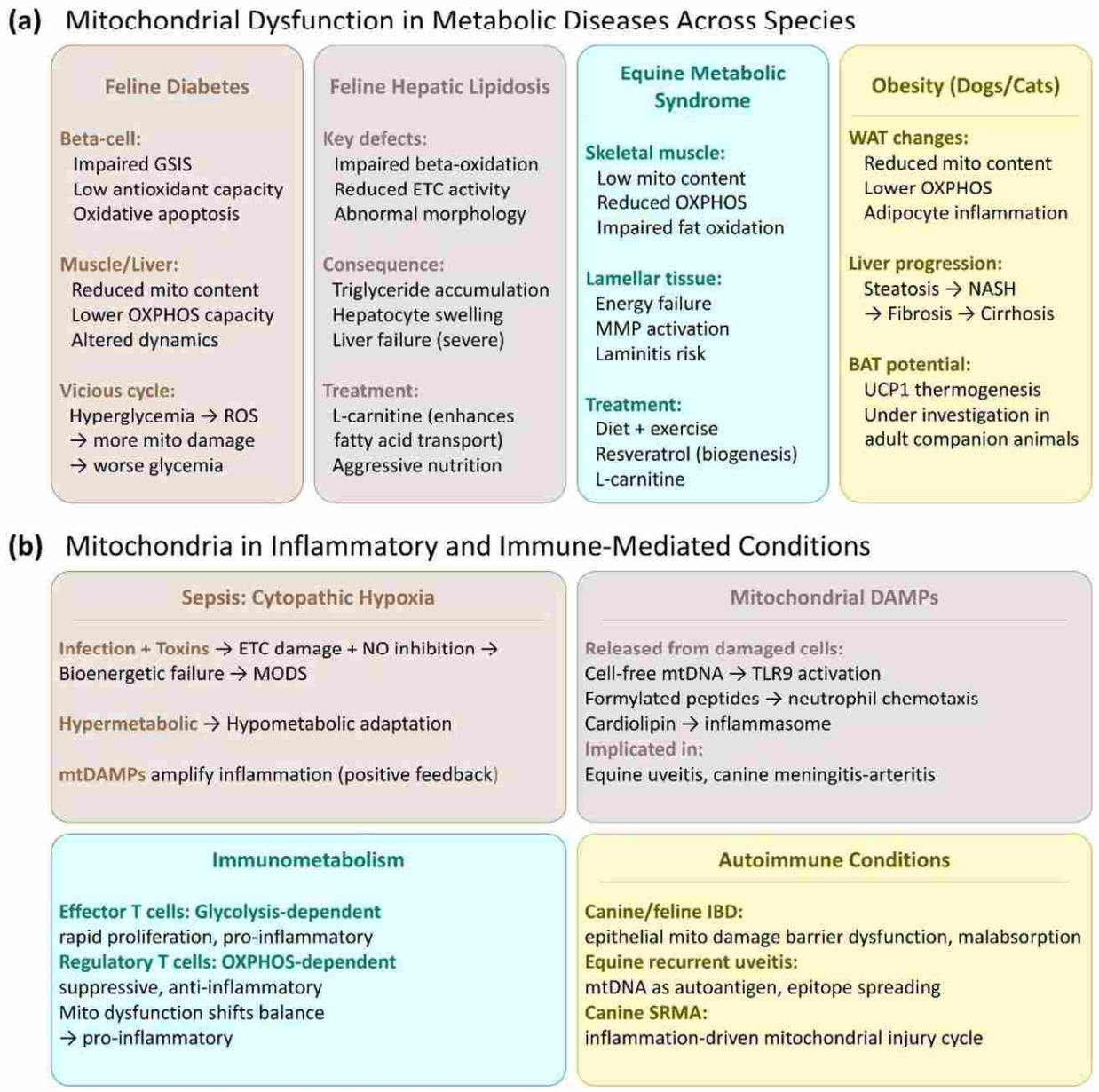

Figure 6(a). Mitochondrial dysfunction in metabolic diseases across veterinary species. *From left to right: feline diabetes mellitus, showing impaired glucose-stimulated insulin secretion in beta cells and reduced mitochondrial content in muscle and liver, with a hyperglycemia-ROS vicious cycle (Zini et al., 2009); feline hepatic lipidosis, featuring impaired beta-oxidation, reduced ETC activity, and triglyceride accumulation overwhelming hepatocyte capacity (Center, 2005); equine metabolic syndrome, with low mitochondrial content in skeletal muscle, lamellar energy failure risking laminitis, and management through diet, exercise, and emerging biogenesis-promoting agents (Frank et al., 2010); and companion animal obesity, showing reduced WAT mitochondrial content, adipocyte inflammation, hepatic steatosis progression, and emerging investigation of BAT thermogenic potential.* ***Figure67(b. Mitochondria in inflammatory and immune-mediated conditions.*** *Upper left: sepsis pathway from infection through ETC damage and bioenergetic failure to multiple organ dysfunction syndrome (MODS), with hypermetabolic-to-hypometabolic phase transition and mtDAMP-mediated inflammatory amplification (Singer, 2014). Upper right: mitochondrial damage-associated molecular patterns (DAMPs), including cell-free mtDNA, formylated peptides, and cardiolipin, activating innate immune receptors. Lower left: immunometabolism, illustrating that effector T cells depend on glycolysis while regulatory T cells rely on OXPHOS, with mitochondrial dysfunction shifting this balance toward pro-inflammatory phenotypes (Michalek et al., 2011). Lower right: autoimmune conditions including canine/feline IBD, equine recurrent uveitis, and canine steroid-responsive meningitis-arteritis, each featuring distinct mitochondrial injury patterns.*

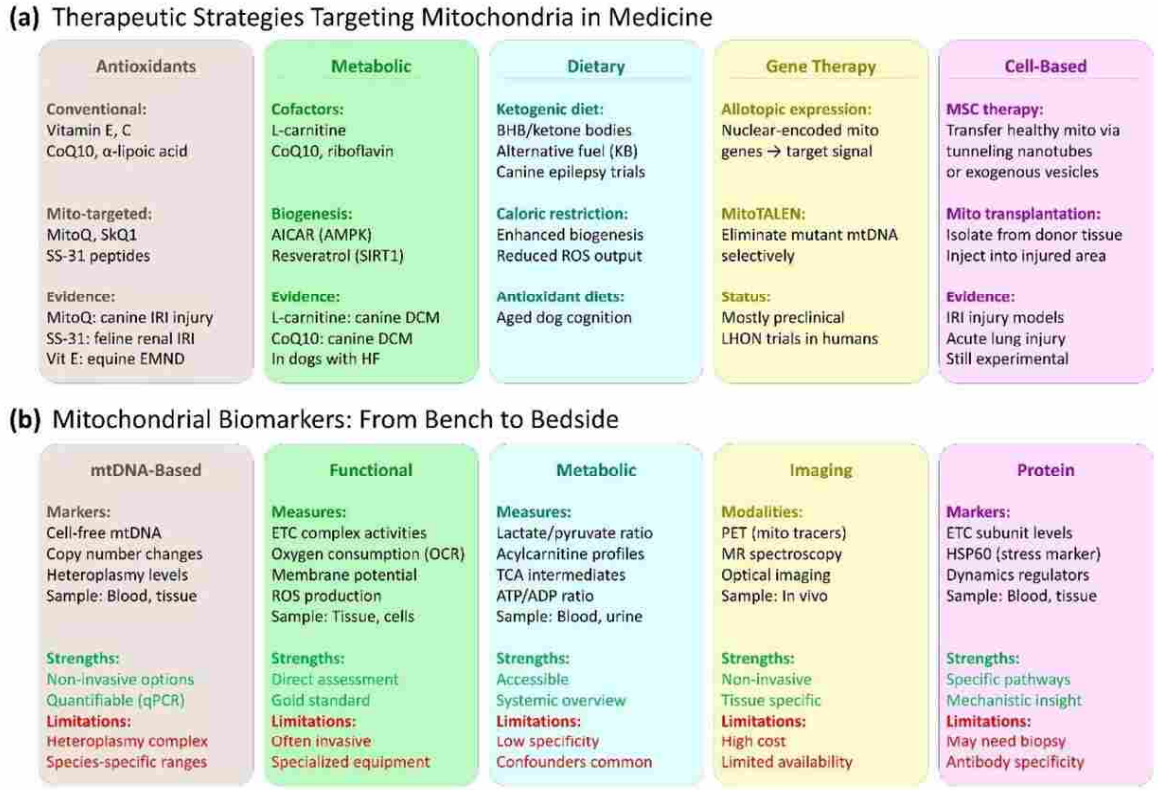

***Figure 7a. Therapeutic strategies targeting mitochondria in veterinary medicine**. Five categories are presented. Antioxidants: conventional (vitamin E, CoQ10) and mitochondria-targeted (MitoQ, SS-31), with evidence from canine IR injury and equine EMND but caution regarding disruption of normal redox signaling. Metabolic modulators: L-carnitine and CoQ10 as cofactors, AICAR and resveratrol as biogenesis enhancers, with clinical evidence primarily from canine cardiac disease. Dietary approaches: ketogenic diet for canine epilepsy, caloric restriction, and antioxidant-enriched diets for aged dogs. Gene therapy: allotopic expression and MitoTALEN for heteroplasmy shifting, currently preclinical. Cell-based therapies: mesenchymal stem cell mitochondrial transfer and direct mitochondrial transplantation, shown in IR injury models but still experimental. Based on Szeto (2014) and Nightingale et al. (2016). **Figure 7b. Mitochondrial biomarkers: from bench to veterinary bedside**. Five biomarker categories are compared. mtDNA-based: cell-free mtDNA, copy number, mutations, and heteroplasmy levels, measurable from blood and tissue with non-invasive options but complicated by heteroplasmy interpretation. Functional: ETC complex activities, oxygen consumption rate, membrane potential, and ROS production, providing direct assessment but often requiring invasive sampling. Metabolic: lactate/pyruvate ratio, acylcarnitine profiles, TCA intermediates, and ATP/ADP ratio, accessible from blood and urine but with limited specificity. Imaging: PET with mitochondrial tracers and MR spectroscopy, non-invasive and tissue-specific but costly and scarce. Protein: ETC subunit levels, HSP60, and dynamics regulators, offering mechanistic insight but often requiring biopsy. Strengths and limitations are listed for each category. Based on Brand and Nicholls (2011) and Tyrrell et al. (2017).*

10 References

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